

PASA

Permit Application Screening Assistant

A Multi-Agent Document Compliance Verification System for Business and Building Permit Applications in Local Government Units



Adrian Salinas
Project Manager



Theodore Samuel Navarro
Lead Developer



Zallen Fritz Eslabra
UI/UX Designer

Bachelor of Science in Information Technology

[University of San Agustin · BSIT - 3C

The defect is found at the counter — after the trip has already been paid for.

3-7-20

Working days within which an LGU must act on simple, complex, and highly technical transactions under RA 11032. The count begins only when the application is complete — so a returned applicant restarts at day zero.

REPUBLIC ACT 11032 • EASE OF DOING BUSINESS ACT (2018)

Who is affected

MSME owners and homeowners filing new and renewal permits at the partner LGU — the majority filing without a fixer or consultant.

What it costs them

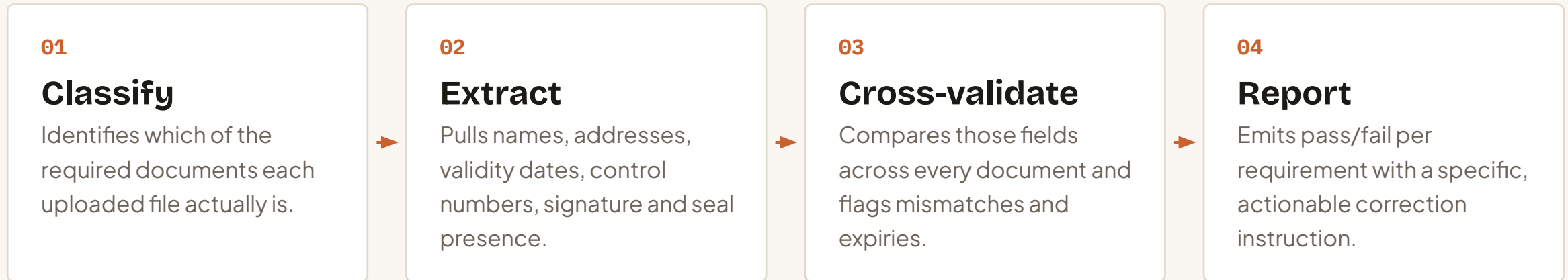
Repeat trips to the municipal hall, transport fare, lost daily wage, and a renewal deadline that keeps moving.

What it costs the LGU

Processor time spent rejecting rather than evaluating, longer counter queues, and ARTA compliance exposure on posted service pledges.

[[TBD: baseline % of applications returned for document defects — to be obtained from partner BPL0 records]]

Screen the documents before the applicant travels, not after they arrive.



What makes it different

- Existing e-permit portals accept an upload; none of them read it. PASA verifies.
- Cross-document consistency — not just presence — is the defect class that actually causes returns.
- Output is a correction instruction, not an error code: what to fix, on which document, and where to get it.
- The same verdict is visible to the processor, so staff review a pre-screened queue.

Uploads are already possible. Verification is not.

	Accepts uploads	Checks each doc for completeness	Cross-document consistency	Feedback before the trip	Legally binding determination
Manual check at the BPLO counter The incumbent process	—	Yes	Yes	No	Yes
eBPLS / LGU online permit portals DICT-supported deployments	Yes	No	No	Partial	Yes
Generic OCR & document scanners Adobe Scan, Google Lens, CamScanner	Yes	No	No	No	No
PASA Proposed	Yes	Yes	Yes	Yes	No

Honest verdict: only the counter check is legally binding — PASA doesn't replace it, just the wasted trip beforehand.

What we will build, and how we will know it worked.

GENERAL OBJECTIVE

To design, develop, and evaluate PASA — a multi-agent system that screens permit requirements for completeness, validity, and cross-document consistency before the applicant submits them.

SPECIFIC OBJECTIVES

- | | | | |
|-----------|---|-----------|---|
| 01 | To design a normalized schema with row-level security for applicant, processor, and approver access. | 02 | To develop a classification and field-extraction pipeline for each uploaded requirement. |
| 03 | To implement a cross-document validator flagging mismatched names, addresses, and expired dates. | 04 | To generate a per-requirement pass/fail report with a specific correction instruction. |
| 05 | To determine detection accuracy — precision, recall, F1 — against licensed BPLO processor judgments. | 06 | To evaluate usability with applicants and staff using an ISO/IEC 25010 instrument. |

Three roles, three surfaces, one requirement engine.

Applicant

PUBLIC USER

- Guided requirement checklist per permit type
- Multi-file upload with re-capture prompts on illegible pages
- Compliance report: pass/fail per requirement
- Correction instructions and where to obtain each document

Processor

BPLO / OBO STAFF

- Pre-screened application queue, sorted by compliance state
- Side-by-side view of source document and extracted fields
- Override and annotate any automated finding
- Log the official determination — the ground-truth record

Approver / Admin

DEPARTMENT HEAD

- Requirement template management per permit type
- Validity-period and field-rules configuration
- Analytics on the most frequent rejection reasons
- Immutable audit log of every access and override

Permit types in scope: new and renewal business permits, and building permits, at one partner LGU.

What this system will not do.

OUT OF SCOPE

Advisory only

PASA does not issue, approve, or deny a permit. The BPLO determination is authoritative.

No authenticity verification

No API integration with DTI, SEC, BFP, or BIR. We detect missing and inconsistent — not forged.

No payment or issuance

Fee assessment, payment, and permit release remain with the LGU's existing system.

DEFERRED TO PHASE 2

Handwritten documents

Pages below the legibility threshold are routed to manual review.

Offline operation

Web-based and online only. No offline capture for low-connectivity barangays.

Single partner LGU

Templates are configurable, but validation is against one LGU's requirement set.

Each deferred item depends on access or data we do not yet have. All are documented as Phase 2 work rather than dropped.

Agile Scrum — because the requirement set is the thing we are least sure about.

Why Scrum fits this project

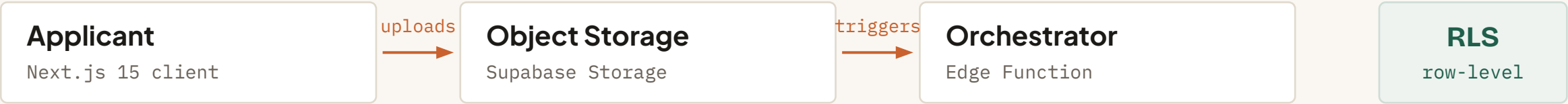
- Requirement templates are unknown until BPLO staff validate them — they will change after Sprint 3.
- Extraction accuracy improves only by iterating against real document samples, one sprint at a time.
- A four-member team with fixed academic deadlines needs a fixed cadence more than it needs a fixed plan.
- Every sprint produces something demonstrable to the adviser, which de-risks the final defense.

Sprint cadence

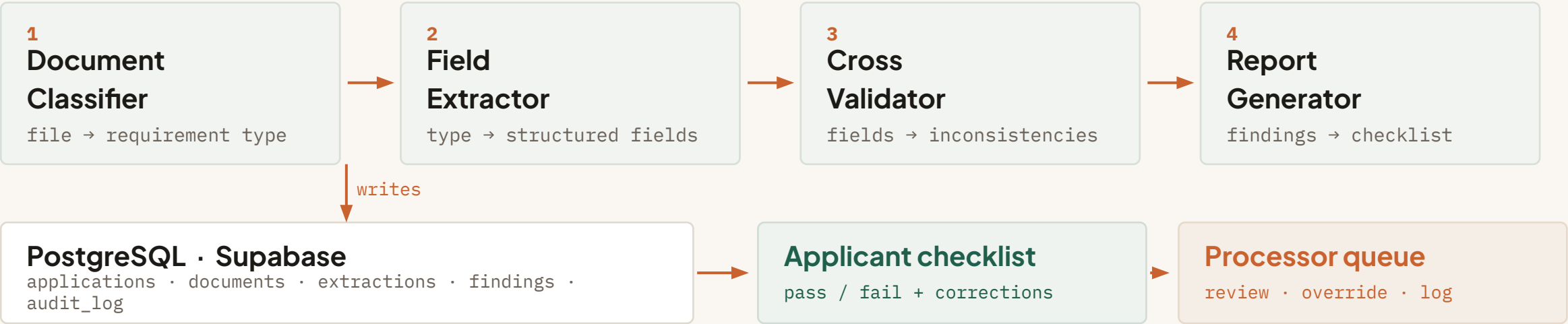
SPRINT LENGTH	2 weeks
PLANNED SPRINTS	8 · [[TBD: confirm dates]]
CEREMONIES	Planning · daily stand-up · review · retrospective
LGU VALIDATION	End of Sprints 3, 5, and 7
DEFINITION OF DONE	Merged · typed · Zod-validated · tested · staged

Requirements → Design → Build → Validate with LGU → Refine → Evaluate → Deploy

Context diagram: a four-agent verification pipeline.

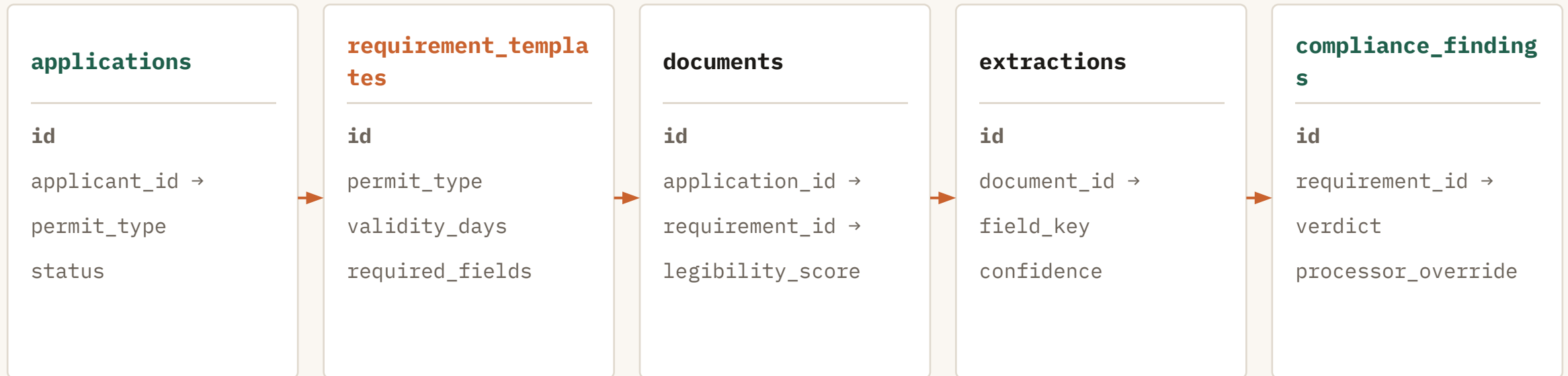


AGENT PIPELINE · each stage passes a Zod-validated object to the next



Full level-1 DFD and ERD: Appendix A1-A2.

Five core entities. Requirements are data, not code.



Why requirement_templates is the keystone

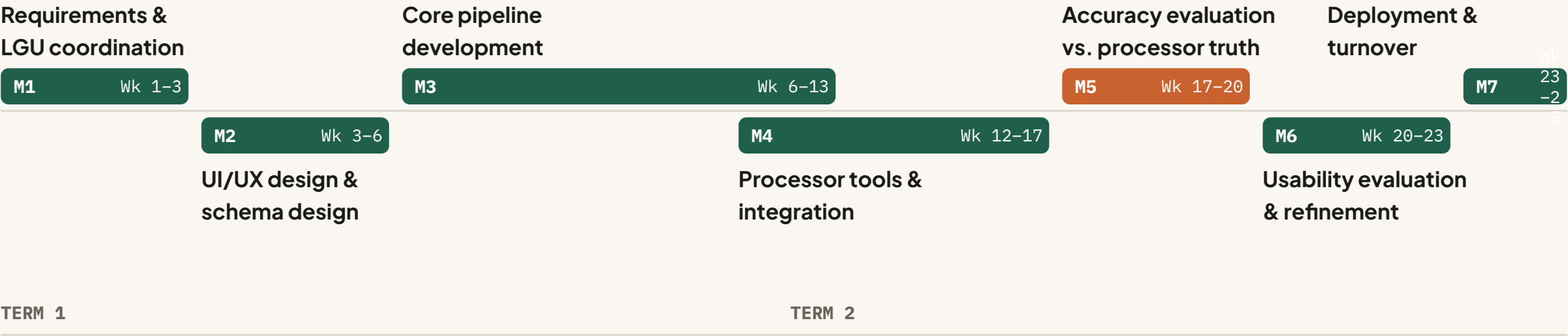
Requirement sets differ by LGU and change by ordinance — storing them as rows makes a new permit type an admin edit, not a redeploy.

Typed at every boundary, authorised at the database.

Frontend	Next.js 15 (App Router) React · TypeScript strict Tailwind CSS · shadcn/ui	Server Components by default; client interactivity only where upload and review require it.
Backend & agents	Supabase Edge Functions (Deno) Vision-capable LLM, JSON-schema constrained Zod validation at every hand-off	Each agent returns structured output parsed against a schema before the next stage runs.
Database & storage	PostgreSQL via Supabase Row-Level Security policies Supabase Storage, signed URLs	Role separation enforced in SQL, not in the client. Documents are never publicly addressable.
Tooling & infrastructure	Vercel · GitHub · GitHub Actions ESLint · Prettier · Vitest Figma for design handoff	CI runs type-check, lint, and tests on every pull request before deploy.

Alternative: Firebase — rejected, since document-model joins would move integrity checks into application code.

Seven milestones. Accuracy is measured before deployment, not after.



M5 precedes M7 on purpose — if accuracy falls short, schedule remains to respond.

Sprint-level Gantt: Appendix A3 · [[TBD: calendar dates]]

How we will prove it works.

MEASURE	METRIC	RESPONDENTS / DATA	DESIGN NOTES
Detection accuracy	Precision · Recall · F1 per requirement type	Real document sets, n = [[TBD]]	Ground truth: 3 BPLO processors, majority resolution. Recall weighted higher — missed defects cost the trip.
Rejection-rate reduction	% of applications returned for defects	BPLO baseline vs. PASA-screened cohort	Pre/post comparison over a matched period. Baseline: [[TBD]].
Processing time	Minutes at the counter; days to complete filing	Timed observation, with system vs. without	Both ends: processor time per application, applicant elapsed days.
Usability & acceptability	Weighted mean per ISO/IEC 25010	Applicants n = [[TBD]] · staff: full enumeration	Expert-validated instrument. Interpretation scale in Appendix A4.

Weighted mean for Likert data; precision, recall, F1 for detection. Sample-size computation: Appendix A4.

Every objective has a feature and a metric.

#	SPECIFIC OBJECTIVE	FEATURE THAT DELIVERS IT	HOW EVALUATED	METRIC
01	Design secure schema with RLS	Role-scoped data access	Policy test suite; unauthorised-access attempts	0 successful cross-role reads
02	Develop classification & extraction	Upload → structured fields	Per-field extraction accuracy on labelled set	Field-level accuracy ≥ [[TBD]]%
03	Implement cross-validator	Name / address / date consistency	Seeded inconsistency test set	Detection rate on seeded defects
04	Generate compliance report	Pass/fail + correction instructions	Applicant comprehension task	% who correctly act on instruction
05	Determine detection accuracy	Whole pipeline	Comparison vs. 3 BPLO processors	Precision · Recall · F1
06	Evaluate usability	Whole system	ISO/IEC 25010 instrument	Weighted mean per characteristic

No objective without a feature. No feature without a metric.

Every document here is personal information.

What is collected

Scanned or photographed requirements — names, addresses, control numbers. Nothing beyond the posted requirement set.

Where it is stored

Private Supabase Storage, single region. Never publicly addressable; access via short-lived signed URLs only.

Who can access it

Applicants see their own applications; processors see only their office's queue. Enforced by RLS in PostgreSQL, not client-side.

Retention & model use

Purged [[TBD]] days after final determination. Never used to train or fine-tune any model — zero-retention inference.

COMPLIANCE POSTURE

RA 10173 (Data Privacy Act) — a processing agreement with the LGU is signed before any live document is collected.

Who benefits, and how.

Permit applicants

MSME OWNERS AND HOMEOWNERS

One trip instead of three — a defect found at home is a same-day fix, not a lost day of income.

LGU processors

BPLO AND OBO FRONT-LINE STAFF

A pre-screened queue shifts counter time from rejecting applications to evaluating them — and shortens the line.

Policy & oversight

LGU MANAGEMENT, ARTA REPORTING

Findings become a record of which requirements fail most — evidence for simplifying the requirement set.

The field

IT AND PUBLIC-SERVICE INFORMATICS

Cross-document validation on photo-captured, low-resource documents is thinly studied and has direct civic use.

Expected output: a deployed, LGU-validated screening system with a documented accuracy evaluation against processor ground truth.

PASA

Screen the documents before the applicant travels.

OBJECTIVES

- Design an RLS-secured schema for applicant, processor, and approver roles
- Develop the document classification and field-extraction pipeline
- Implement cross-document consistency validation
- Generate per-requirement pass/fail reports with correction instructions
- Determine detection accuracy against BPLO processor ground truth
- Evaluate usability with applicants and staff (ISO/IEC 25010)

EXPECTED OUTPUT

A deployed compliance screening system for business and building permits, validated at one partner LGU.

Detection accuracy	Precision · Recall · F1
Rejection rate	Pre vs. post baseline
Usability	ISO/IEC 25010 weighted mean

We are asking the panel to approve this scope.

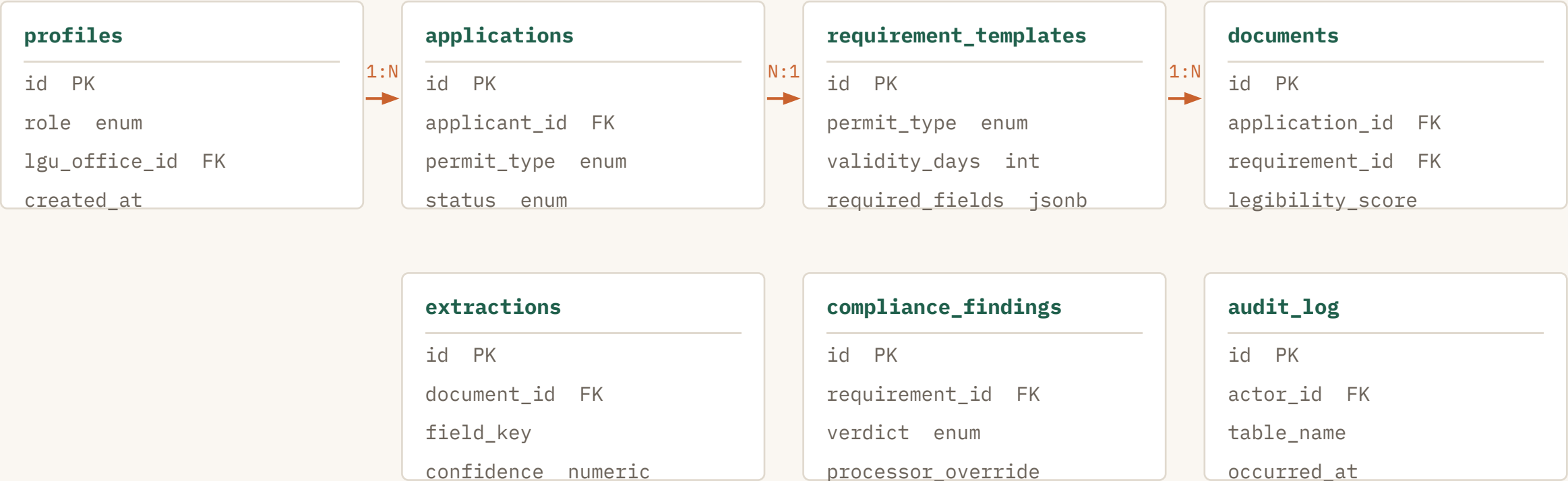
Questions welcome.

Level-1 data flow diagram



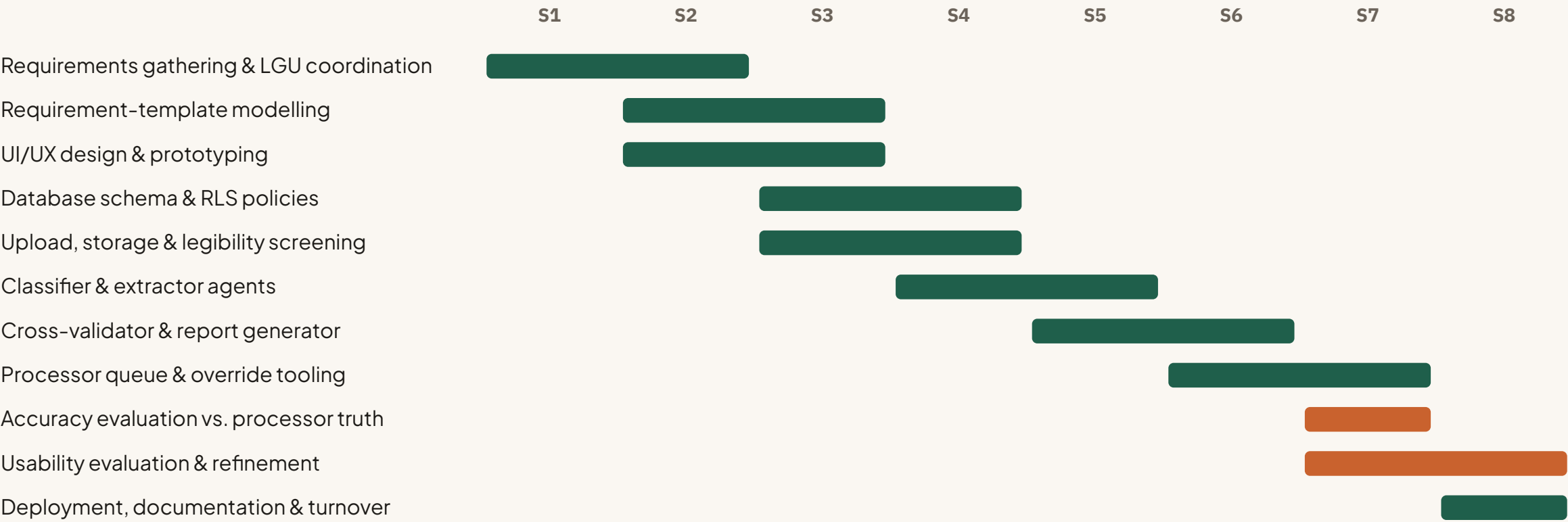
Flow runs strictly left to right. Process 1.0 rejects illegible pages before any inference cost is incurred; only 4.0 reads more than one document at a time.

Entity relationship diagram – full



profiles 1:N applications · applications 1:N documents · documents 1:N extractions · applications 1:N compliance_findings · audit_log references all

Sprint-level Gantt chart



LGU validation sessions at the end of S3, S5, and S7 · [[TBD: calendar dates]]

Sampling, instrument, and interpretation scale

RESPONDENTS

- BPLO / OBO staff — whole-population enumeration. Population is small, known, and fully reachable; sampling would discard data.
- Applicants — computed sample from the partner LGU's monthly application volume, [[TBD: N]], via Slovin's formula at 5% margin of error.
- Ground-truth panel — 3 licensed processors, independent judgment, majority resolution on disagreement.

INSTRUMENT

- Usability survey adapted from ISO/IEC 25010 quality characteristics: functional suitability, usability, reliability, security, maintainability.
- Expert validation by 3 evaluators prior to deployment; items revised on their consolidated comments.
- Separate short instrument for applicants, written in Filipino and English.

ANALYSIS

- Likert items: weighted mean, interpreted against the scale below.
- Detection performance: precision, recall, F1 per requirement type; confusion matrix reported in full.
- Rejection rate and processing time: descriptive comparison of pre- and post-intervention cohorts.

INTERPRETATION SCALE

4.21 – 5.00 Highly Acceptable	3.41 – 4.20 Acceptable	2.61 – 3.40 Moderately Acceptable	1.81 – 2.60 Slightly Acceptable	1.00 – 1.80 Not Acceptable
---	----------------------------------	---	---	--------------------------------------

Alternative titles considered

CHOSEN	PASA: Permit Application Screening Assistant	Names the mechanism and the beneficiary — Filipino "pasa" (to pass, to hand over) carries the outcome applicants want.
ALTERNATIVE	DOKUMENTO: A Cross–Document Consistency Validator for LGU Permit Requirements	Foregrounds the technical contribution over the service — stronger for a technically inclined panel.
ALTERNATIVE	HANDA: Pre–Submission Compliance Verification for Business and Building Permits	Emphasises readiness and the anti-red-tape framing. Aligns closely with RA 11032 language.
ALTERNATIVE	An Agent–Based Document Verification System for Local Government Permit Processing	Conventional academic phrasing. Safest, least memorable, and gives no signal about the beneficiary.

Scope is unchanged across all four. Only the framing differs.

Operating cost, risk, and turnover

COST CENTRE	BASIS	ESTIMATE
Hosting & database	Supabase + Vercel free tier during development; paid tier at production volume	[[TBD: ₱/month]]
Model inference	Per-document cost; the dominant variable expense. Legibility gate rejects unusable pages before any paid call.	[[TBD: ₱/application]]
Storage	Documents purged on the retention schedule, which caps steady-state storage cost.	[[TBD: ₱/month]]
Maintenance	Repository, deployment guide, and admin manual turned over to the partner LGU's IT focal person.	Turnover at M7

PRINCIPAL RISKS

LGU access not granted

Fallback: synthetic and public specimen documents, study reframed as a controlled-set evaluation.

Extraction accuracy below target

M5 precedes deployment specifically so the schedule absorbs a remediation sprint.

Requirement set changes mid-project

Templates are data; an ordinance change is an admin edit, not a redeploy.